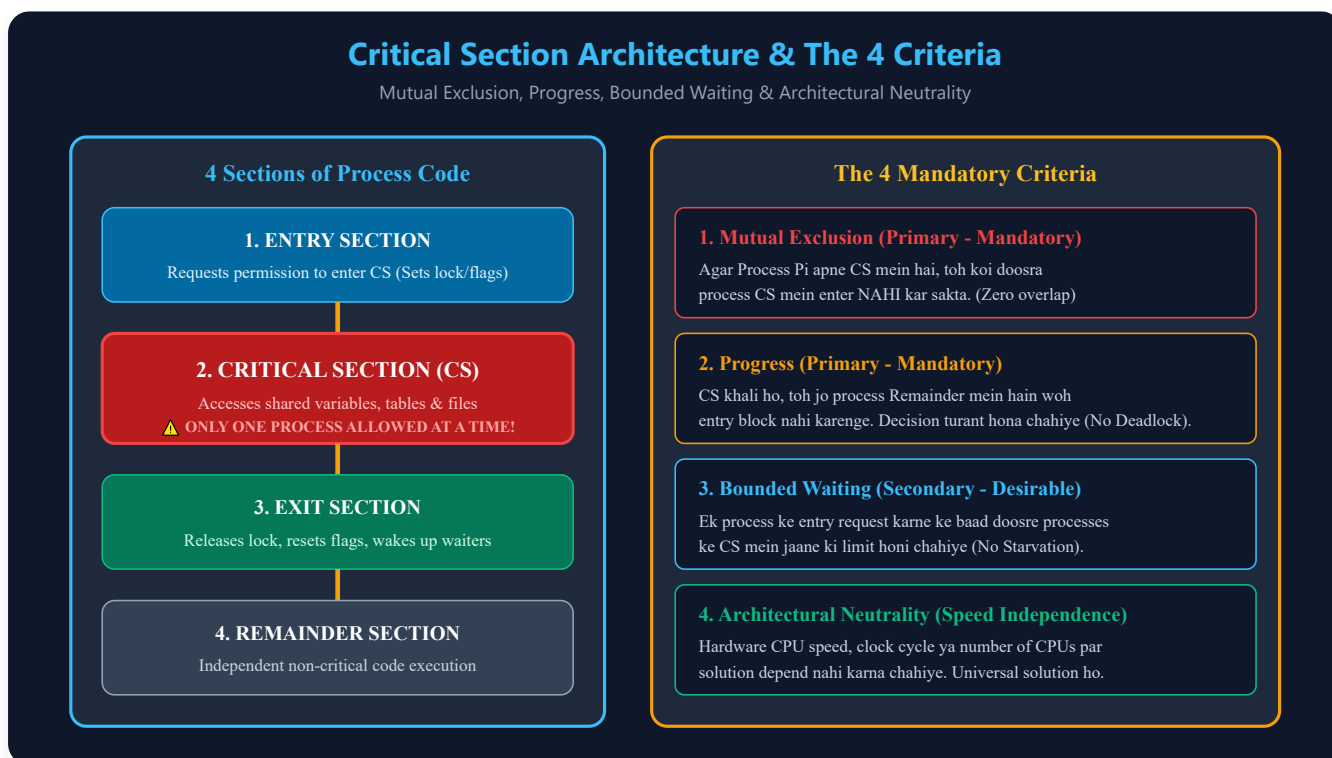


2. Critical Section Problem & The 4 Criteria

Syllabus Coverage: Gateway Classes Slides 16–24 | 4 Sections of Process Code, Primary vs Secondary Criteria.

2.1 The 4 Sections of Process Code

- **Entry Section:** Lock acquire karne aur CS mein ghusne ki permission ka code.
- **Critical Section (CS):** Shared data manipulation code (Only ONE process allowed).
- **Exit Section:** Lock release aur waiting processes ko inform karne ka code.
- **Remainder Section:** Baki ka normal independent code execution.



2.2 The 4 Solution Criteria

Criterion	Type	Core Definition	Prevention Goal
Mutual Exclusion	Primary (Mandatory)	Only one process can execute in CS at any given time.	Data Inconsistency
Progress	Primary (Mandatory)	Selection of next process cannot be blocked by remainder processes or postponed indefinitely.	Deadlock
Bounded Waiting	Secondary (Desirable)	Bound on the number of entries granted to others after a request is made.	Starvation
Architectural Neutrality	Secondary (Desirable)	No assumptions regarding relative CPU speeds or hardware clocks.	Portability bugs

2.3 Common Interview Traps

Trap: Kya Bounded Waiting satisfy na hone par solution reject ho jata hai?

Answer: Primary conditions (Mutual Exclusion + Progress) strictly mandatory hain. Agar Bounded Waiting satisfy na ho, toh solution partial mana jata hai (starvation prone), lekin system deadlock me nahi fasata.